DSA03-Natural Language Processing

1.Write program demonstrates how to use regular expressions in Python to match and search for patterns in text.

Aim: To write a Python program to match and search for patterns in text using regular expressions.

Procedure:

1. Import the re module.
2. Define the input text.
3. Search for a pattern using re.search().
4. Find matching numbers using re.findall().
5. Display the output.

Code:

```
import retext = "My phone number is 9876543210"
result = re.search(r"\d{10}", text)
print("Search result:", result.group())
numbers = re.findall(r"\d+", text)
print("All numbers:", numbers)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/1.py"
Search result: 9876543210
All numbers: ['9876543210']

C:\Users\blohi\Downloads\NLP lab>
```

Result: Regular expressions were used successfully to search and match patterns in text.